



# ACAVCaps: Enabling Large-Scale Training for Fine-Grained and Diverse Audio Understanding



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## Highlights

- **Motivation:** Existing audio captioning datasets are limited in scale, lexical diversity, and domain coverage, hindering the development of generalizable audio understanding models.
- **Solution:** ACAVCaps is introduced — a large-scale, fine-grained audio captioning dataset derived from ACAV100M.
- **Advantages:**
  - **Data Source:** A novel data source (ACAV100M), entirely distinct from existing captioning datasets (e.g., AudioSet).
  - **Pipeline:** Multi-expert analysis + LLM Chain-of-Thought reasoning for rich, multi-faceted captions.
  - **Audio Coverage:** Diverse audio domains, including speech, music, sound, and their various combinations.
  - **Text Coverage:** Multi-perspective captions spanning systematic, content-related, and content-unrelated dimensions.

## Dataset Comparison

ACAVCaps is compared against existing audio captioning datasets in terms of scale, source diversity, and descriptive richness.

- **Source:** ACAV100M Versus AudioSet, FreeSond and Others
- **Scale:** **4.7M** audio-text pairs spanning **13,000 hours**
- **Richness:** **76.7K** unique tokens, far exceeding Combined<sup>†</sup> (47.6K)

| Labeling | Dataset                       | Samples     | Unique Tokens | Domain                   | Source          |
|----------|-------------------------------|-------------|---------------|--------------------------|-----------------|
| Manual   | AudioCaps                     | 50k         | 5.5k          | Multi-Domain             | AudioSet        |
|          | Clotho                        | 3.8k        | 5.5k          | Multi-Domain             | FreeSound       |
|          | SongDescriber                 | 0.4k        | 2.4k          | Music                    | MTG-Jamendo     |
| LLM      | MusicCaps                     | 4.6k        | 4.6k          | Music                    | AudioSet        |
|          | LPMusicCaps                   | 21.6k       | 5.1k          | Music                    | AudioSet, MSD   |
|          | WavCaps                       | 0.4M        | 23.1k         | Multi-Domain             | AudioSet        |
|          | Auto-ACD                      | 1.9M        | 20.3k         | Multi-Domain             | AudioSet        |
|          | Sound-VeCaps                  | 1.6M        | 42.7k         | Multi-Domain             | AudioSet        |
|          | AudioSetCaps                  | 2.0M        | 20.9k         | Multi-Domain             | AudioSet        |
|          | MP-LLM <b>ACAVCaps (Ours)</b> | <b>4.7M</b> | <b>76.7k</b>  | <b>Ext. Multi-Domain</b> | <b>ACAV100M</b> |

## Downstream Task Generalization

Models pre-trained on each captioning dataset are evaluated on diverse downstream tasks to assess how dataset quality translates to generalizable representations.

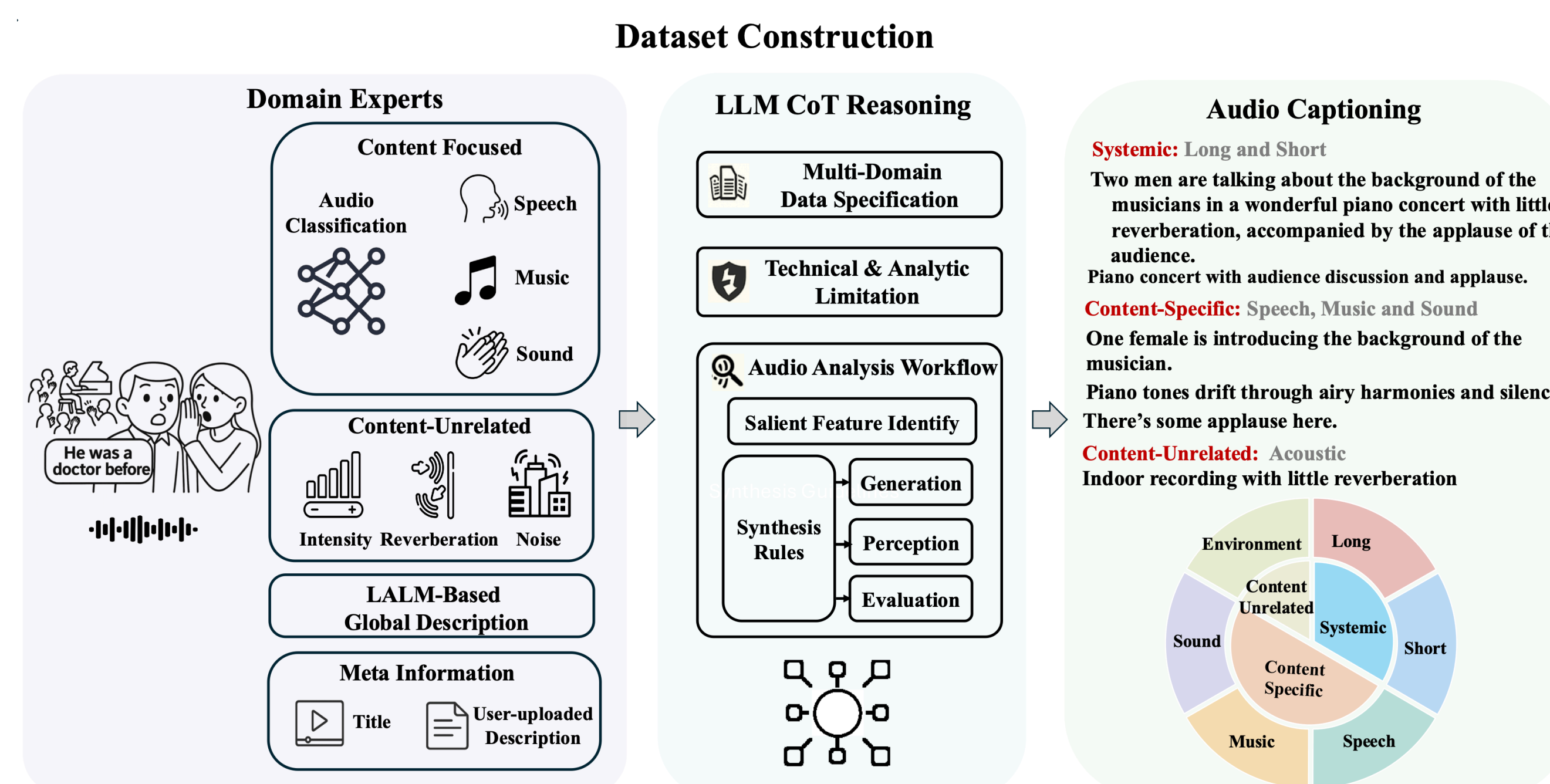
- **Training:** Frozen Audio encoder and LLM; only the MLP adapter is fine-tuned
- **Tasks:** Speech, Sound, Music and Paralanguage
- **Results:** Superior or competitive performance is achieved across all domains

| Training Dataset      | Speech ↓    |             |             |             |              |             | Sound ↑     |             | Music ↑     | Other ↑     |
|-----------------------|-------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|-------------|-------------|
|                       | AISHELL-2   |             | LibriSpeech |             | Common Voice | General     | Vocal       | Instrument  | Emotion     |             |
|                       | Android     | IOS         | MIC         | Clean       |              |             |             |             |             | Other       |
|                       |             |             |             |             |              |             | VGGSound    | VocalSnd    | NSynth      | IEMOCAP     |
| AudioSetCaps          | 82.7        | 77.8        | 81.7        | 51.6        | 70.2         | 84.7        | 22.4        | 91.4        | 67.0        | 17.6        |
| Auto-ACD              | 89.1        | 78.2        | 88.6        | 54.6        | 76.5         | 85.7        | 22.5        | 90.2        | 46.1        | 24.1        |
| WavCaps               | 83.2        | 74.2        | 77.9        | 54.3        | 74.0         | 85.2        | 21.2        | 91.5        | <b>69.1</b> | 19.9        |
| Sound-VeCaps          | 87.3        | 79.5        | 87.9        | 51.8        | 70.1         | 85.6        | 22.9        | 90.8        | 45.0        | 20.3        |
| Combined <sup>†</sup> | 84.2        | 76.4        | 82.3        | 41.5        | 59.4         | 83.0        | <b>34.6</b> | <b>92.6</b> | 44.0        | 19.8        |
| <b>ACAVCaps</b>       | <b>58.3</b> | <b>56.5</b> | <b>57.1</b> | <b>19.7</b> | <b>33.7</b>  | <b>50.0</b> | 20.4        | 92.1        | 64.7        | <b>28.9</b> |

<sup>†</sup>Combined: AudioSetCaps, Auto-ACD, WavCaps, Sound-VeCaps

## Data Construction Framework

Audio is analyzed by a **multi-expert pipeline** across multiple perspectives, then synthesized via **LLM Chain-of-Thought reasoning** into rich, **multi-faceted captions**.



## Audio Captioning Performance

The quality of audio captioning is assessed on the MECAT-Caption benchmark using the DATE score.

- **Training:** Audio encoder, adapter, and LLM (LoRA) are jointly trained
- **Tasks:** Systematic, content-related and content-unrelated of MECAT-Caption
- **Results:** Superior performance is achieved across **all** sub-tasks

| Training Dataset       | Systematic  |             | Content-Related |             |             |             |             |             | Content-Unrel. | Score       |
|------------------------|-------------|-------------|-----------------|-------------|-------------|-------------|-------------|-------------|----------------|-------------|
|                        | Long Short  |             | Speech          |             | Music       |             | Sound       |             | Environ.       |             |
|                        | Pure        | Mixed       | Pure            | Mixed       | Pure        | Mixed       | Pure        | Mixed       |                |             |
| AudioSetCaps           | 52.4        | 52.0        | 30.2            | 31.4        | 44.3        | 30.9        | 52.4        | 21.6        | 15.4           | 37.4        |
| Auto-ACD               | 47.3        | 50.0        | 29.1            | 31.0        | 26.9        | 21.9        | 49.5        | 18.9        | 11.0           | 32.8        |
| WavCaps                | 47.3        | 50.9        | 27.3            | 30.1        | 15.9        | 19.4        | 46.5        | 20.0        | 9.2            | 31.4        |
| Sound-VeCaps           | 47.0        | 49.7        | 29.1            | 30.3        | 27.2        | 21.9        | 49.8        | 18.7        | 11.4           | 32.8        |
| Combined <sup>†</sup>  | 52.2        | 54.1        | 30.2            | 32.2        | 45.4        | 23.3        | 52.7        | 20.2        | 11.1           | 36.6        |
| <b>ACAVCaps (Ours)</b> | <b>76.6</b> | <b>75.7</b> | <b>64.2</b>     | <b>64.9</b> | <b>60.5</b> | <b>41.1</b> | <b>59.5</b> | <b>28.0</b> | <b>34.8</b>    | <b>60.9</b> |

<sup>†</sup>Combined: AudioSetCaps, Auto-ACD, WavCaps, Sound-VeCaps.

## Conclusion

To address the limitations of existing audio captioning datasets in scale, scope, and descriptive detail, ACAVCaps is proposed. Experiments confirm that **large-scale, comprehensive, and richly described data** is crucial for robust audio representations, with superior captioning and strong generalization to downstream tasks.



[github.com/xiaomi-research/acavcaps](https://github.com/xiaomi-research/acavcaps)